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A modified Mediterranean dietary intervention for adults with major depression: Dietary protocol and feasibility data from the SMILES trial

Rachelle S. Opie¹, Adrienne O'Neil^{2,3}, Felice N. Jacka ^{2,4,5,6}, Josephine Pizzinga², Catherine Itsiopoulos¹

¹School of Allied Health, La Trobe University, Bundoora, VIC, Australia, ²Food and Mood Centre, IMPACT SRC, Deakin University, Geelong, VIC, Australia, ³Melbourne School of Population and Global Health, The University of Melbourne, Parkville, VIC, Australia, ⁴Department of Psychiatry, University of Melbourne, Parkville, VIC, Australia, ⁵Centre for Adolescent Health, Murdoch Childrens Research Institute, Parkville, VIC, Australia, ⁶Black Dog Institute, Randwick, NSW, Australia

Background: The SMILES trial was the first randomized controlled trial (RCT) explicitly designed to evaluate a dietary intervention, conducted by qualified dietitians, for reducing depressive symptomatology in adults with clinical depression.

Objectives: Here we detail the development of the prescribed diet (modified Mediterranean diet (Mod/MedDiet)) for individuals with major depressive disorders (MDDs) that was designed specifically for the SMILES trial. We also present data demonstrating the extent to which this intervention achieved improvements in diet quality.

Methods: The Mod/MedDiet was designed using a combination of existing dietary guidelines and scientific evidence from the emerging field of nutritional psychiatric epidemiology. Sixty-seven community dwelling individuals (Melbourne, Australia) aged 18 years or over, with current poor quality diets, and MDDs were enrolled into the SMILES trial. A retention rate of 93.9 and 73.5% was observed for the dietary intervention and social support control group, respectively. The dietary intervention (Mod/MedDiet) consisted of seven individual nutrition counselling sessions delivered by a qualified dietitian. The control condition comprised a social support protocol matched to the same visit schedule and length.

Results: This manuscript details the first prescriptive individualized dietary intervention delivered by dietitians for adults with major depression. Significant improvements in dietary quality were observed among individuals randomized to the Mod/MedDiet group. These dietary improvements were also found to be associated with changes in depressive symptoms.

Discussion/Conclusion: The Mod/MedDiet, a novel and individually tailored intervention designed specifically for adults with major depression, can be effectively implemented in clinical practice to manage this highly prevalent and debilitating condition.

Trial registration: Australia and New Zealand Clinical Trials Register (ANZCTR): ACTRN12612000251820. Registered 29 February 2012

Keywords: Depression, Modified Mediterranean diet, Dietary intervention, Randomized controlled trial, Dietitian

Introduction

Depressive disorders are one of the leading causes of disease burden globally.^{1,2} In many Western countries like Australia, one in five individuals aged 16–85 years have a mental disorder (anxiety, mood, or substance use disorders), and almost half (45% or 7.3 million people) have experienced a mental disorder at some

point in their lives.³ While remarkable advances have been made in the detection and management of depressive disorders over the past decade,⁴ the high worldwide prevalence of depression in both developed and developing countries^{5,6} continues to pose challenges for individuals, families, clinicians, and society. Hence, there is a need to investigate new approaches to prevent and better manage depression.

Observational evidence supports the protective role of a healthy dietary pattern, such as the Mediterranean

Correspondence to: Rachelle S. Opie, School of Allied Health, La Trobe University, VIC, Australia.
Email: r.opie@latrobe.edu.au; rachellesopie@gmail.com

diet (rich in fresh fruits, vegetables, fish, extra virgin olive oil, and whole grains) on depression risk.^{7–9} A high intake of ‘processed foods’, consisting of sweetened desserts, fried food, processed meats, and refined cereals, is associated with increased risk, suggestive of deleterious effects.^{10,11} However, the existing data are largely drawn from observational studies, which pose limitations in determining causality. This is of particular concern when considering that the relationship between depression and diet is bi-directional; individuals with depression are more likely to consume poorer quality diets than those without.⁹ For example, clinical depression and depressive symptoms are often associated with altered appetite, which can involve either an increased appetite with cravings or a reduced interest in food.^{12,13} Negative mood states can stimulate a preferential desire to consume salty, sweet, or high-fat foods.^{12,14} Importantly, depression is also commonly associated with fatigue and apathy,¹⁵ which may have impact on an individual’s motivation to engage in healthy dietary habits,¹⁶ a reduced desire to cook,¹⁷ and depleted energy for activities such as grocery shopping and meal preparation.^{18,19} As a result of decreased concentration, decreased mental endurance, and slowed thinking¹⁵ individuals may also find learning new recipes, developing cooking skills, or performing meal preparation challenging.^{18,19}

Finally, depression has a strong bidirectional relationship with other common, chronic diseases with a dietary aetiology, including heart disease, obesity, and diabetes. These comorbidities can complicate treatment approaches and have a detrimental impact on treatment outcomes.²⁰ Thus, for a number of reasons, dietary improvement in individuals with depression is a potentially important therapeutic approach that is likely to yield a range of health benefits.

The SMILES trial, ‘Supporting the Modification of lifestyle In Lowered Emotional States’, was the first randomized controlled trial (RCT) explicitly designed to evaluate a dietary improvement intervention, conducted by Accredited Practising Dietitians, for reducing depressive symptomatology.²¹ At 3 months, large between-group effect sizes of -1.16 were observed in favour of the dietary intervention group. Here we detail the development of the prescribed diet intervention (the modified Mediterranean diet), and the extent to which the diet is feasible and able to achieve improvements in dietary quality.

Methods

Overview of the SMILES study

Detailed study methods of the SMILES trial have been published previously.²² Briefly, the SMILES trial was a 12-week, parallel group, single-blind, RCT of a dietary intervention for the treatment of moderate-to-severe

depression. The study was conducted over two sites: an inner city setting (Melbourne, Australia) and a regional setting (Geelong, Australia). Participants were randomized to receive either the dietary intervention or a control condition (social support ‘befriending’²³). Recruitment and intervention delivery occurred from October 2012 to July 2015.

Participants’ eligibility

Eligible participants were aged 18 years or over, successfully fulfilled the Diagnostic and Statistical Manual of Mental Disorders 4 (DSM-IV-TR) criteria for major depressive disorder (MDD) Single Episode or Recurrent, scored 18 or over on the Montgomery–Åsberg Depression Rating Scale (MADRS),²⁴ and had current poor quality diets as determined by a score of 75 or less (out of 104) in a Dietary Screening Tool.²⁵ Broadly defined, participants had to report a poor (low) intake of dietary fibre, lean proteins, fruits and vegetables, and a high intake of sweets, processed meats, and salty snacks.²⁵ Participants were eligible if they were able to eat foods as prescribed, without religious, medical, or socio-cultural factors precluding participation or adherence to the prescribed diet. If participants were on antidepressant therapy or undergoing psychotherapy, they were required to be on the same treatment for at least 2 weeks prior to randomization.

Intervention intensity

Diet intervention participants received seven individual face-to-face sessions (~60 minutes each) held at the study centre over a 3-month period. The first three sessions occurred weekly and the remaining four sessions occurred fortnightly. Assessments were completed prior to programme commencement (baseline), at programme completion (3 months), and at 6 months following baseline assessment. The control condition (social support) used the same visit schedule and length as the diet intervention in an individualized setting.

Social support control group

The control condition (social support) comprised a befriending protocol.²³ Befriending consisted of a trained person (research assistant (RA)) discussing neutral topics of interest to the participant with the intention of keeping the participant engaged and positive, without employing techniques specifically used in the major models of psychotherapy.²³ Individuals randomized to the control group were not provided with any dietary advice.

Dietary assessment tools

As part of the comprehensive dietetic consult, the dietitian conducted a diet history to assess habitual dietary intake at the first session (diet intervention

group only). This incorporated a checklist of commonly consumed foods to prompt and cross-check accuracy of reporting. Food models and metric measuring utensils were used to assist with the estimation of portion sizes. Information on eating behaviours; dieting and weight history; motivators, barriers, and ambivalence to change were also discussed. Additionally, participants in both groups were asked to complete a 7-day food diary immediately prior to baseline assessment. Participants were encouraged to record their intake at the time of consumption to minimize error. With the participant present, qualified dietitians (intervention group) or trained RAs (control group) examined the food diaries in detail to check for completeness and accuracy. Portion sizes were checked using food models and household measures. The baseline diet history and 7-day food diary were utilized in dietetic consults (intervention group only) to help promote adherence to the modified Mediterranean diet (ModiMedDiet).

The information on food and fluid intake from 7-day food diaries was entered into FoodWorks7 (Ausnut 2007),²⁶ and analysed for nutrient content. Dietary supplement usage was captured in a lifestyle questionnaire. However, this information was not included in the nutrient analysis. Therefore, findings are reflective of food and beverage consumption exclusively. Of note, at baseline and 3-month assessment, there were no statistically significant differences between groups with regard to nutritional supplement usage ($P > 0.05$).

Under- and over-reporting were measured using the Goldberg cut-off²⁷ to provide an indicator of the likely validity of the food diary information. This approach is a validated method that directly compares energy intake (EI) with energy expenditure. Under- and over-reporting were based on the relationship between EI and Estimated Energy Requirements (EER). EI was calculated based on reported intake from 7-day food diaries, and the Schofield equations²⁸ were used to calculate EER. Under-reporters were defined as $EI:EER < 0.76$, acceptable reporters defined as $EI:EER 0.76-1.24$, and over-reporters defined as $EI:EER > 1.24$.²⁷ Under- and over-reporters were not excluded from the analysis, but their impact on findings was explored.

The dietary intervention

The dietary intervention comprised personalized dietary advice and nutrition counselling support, including motivational interviewing, goal setting, and mindful eating, from a qualified dietitian in order to enhance dietary quality and achieve optimal adherence to the recommended diet. This comprised the 'ModiMedDiet', developed by Rachelle S. Opie and Catherine Itsiopoulos, which is discussed in detail below.

Intervention development

Core components

The Mediterranean diet is the most extensively researched diet in the world. In 2014, Sofi *et al.*,^{29,30} published a systematic review and meta-analysis of cohort studies showing that high adherence to the Mediterranean diet significantly reduced the risk of overall mortality, cardiovascular incidence or mortality, cancer incidence or mortality, and incidence of neurodegenerative diseases. Drawing on this, as well as the meta-analysis demonstrating a reduced risk for depression for those with higher adherence to a Mediterranean diet,³¹ and evidence from the emerging field of nutritional psychiatry,³² we created a 'modified' Mediterranean diet (ModiMedDiet) for the needs of our patient population.

Framework

The ModiMedDiet was primarily constructed using the Dietary Guidelines for Adults in Greece³³ and the Mediterranean-type diet principles from the PREDIMED study.³⁴ Additionally, the Australian Dietary Guidelines³⁵ were utilized for relevance in a non-Mediterranean population and for application to the Australian context. These additional foods (e.g. breakfast cereals, whole grain biscuits, and game meats such as kangaroo) are commonly consumed in Australia, and were grouped into relevant categories based on their similar macro- and micro-nutrient compositions. Moreover, Australian Government's endorsed guidelines relevant to cardiovascular disease health were used due to the strong overlap between depression and cardio-metabolic conditions.³⁶⁻³⁸ Finally, publications by Itsiopoulos *et al.*,³⁹ and Kouris-Blazos *et al.*,^{8,31,40,41} were utilized to ensure that the diet closely resembled the nutrient profile of the *traditional* Mediterranean diet.

Like the *traditional* Mediterranean diet,^{34,42} the ModiMedDiet was designed to be rich in vegetables, fruits, and wholegrain cereals with an emphasis on increased consumption of oily fish, legumes, raw-unsalted nuts and seeds, and extra virgin olive oil (as the main source of added fat). A moderate consumption of reduced fat natural dairy products was recommended to limit saturated fat intake in an effort to achieve the nutrient profile of a *traditional* Mediterranean diet. The diet was also modified to include a moderate consumption of lean red meat. This recommendation was based on epidemiologic evidence from nutritional psychiatry showing that women consuming less (or more) than the recommended intake of red meat were more likely to have clinical depressive and/or anxiety disorders, than those consuming the recommended amount.⁴³ Moreover, red meat is a rich source of iron, zinc, and vitamin B12, which are believed to play a potentially protective

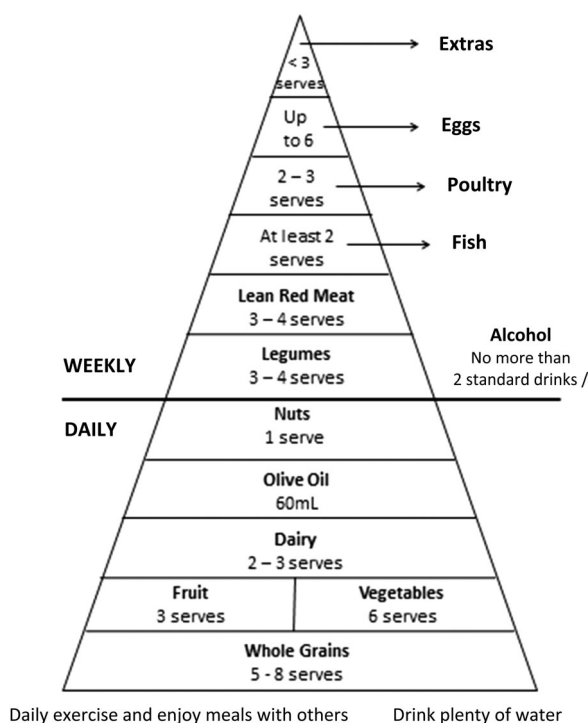


Figure 1 ModiMedDiet food pyramid.

role in common mental disorders.⁴⁴ Promoting adequate lean red meat intake is of further relevance when considering that 38% of Australian women aged 19–50 years have inadequate intakes of iron.⁴⁵

Figure 1 and Table 1 display the 12 food groups, and their corresponding serving sizes, that comprise the ModiMedDiet. Individuals were advised to consume the diet *ad libitum* (e.g. participants were encouraged to eat according to appetite), as the diet intervention did not have a weight loss focus. The ModiMedDiet provided ~11 MJ of daily energy, and was high in fat (40% of energy; >50% from monounsaturated fats), low to moderate in carbohydrate (36% of energy), moderate in protein (18% of energy), and moderate in alcohol (2% of energy from red wine). Based on the nutrient analysis for complete dietary adherence,²⁶ the ModiMedDiet was high in fibre (50 g/d), folate (709 µg/d), and vitamin C (247 mg/d) consistent with large quantities of fruits (411 g/d) and vegetables (709 g/d). Importantly, the ModiMedDiet met all Recommended Dietary Intakes (RDI) or Adequate Intake (AI) requirements for adult males and females,³⁶ with the exception of Vitamin D (Table 2).

Resources and nutrition topics

Participants were provided with a ModiMedDiet resource kit, which included supporting written information designed especially for the intervention in order to enhance dietary compliance. This resource kit included ModiMedDiet dietary guidelines, convenient meal ideas, healthy snack options, shopping lists, and meal plans. The semi-quantitative

ModiMedDiet food pyramid (Fig. 1) was developed using relevant literature from the field of Nutritional Psychiatric Epidemiology, the National Dietary Guidelines and published Mediterranean diet models,^{33,35} and was designed as a simplified and practical visual to assist participants in implementing the dietary intervention.

The material provided was intended to stimulate informed discussions about relevant nutritional topics between the participant and the dietitian. Refer to Supplementary Material for an example of the ModiMedDiet resource kit. Participants were encouraged to set personalized goals at each session and identify barriers/motivators to change. Where relevant, follow-up sessions incorporated themes to support behavioural change, such as label reading, recipe modification, time-management, and food security (access to healthy food and tips for budget buying). At the final session, goals achieved during the intervention period were discussed and summarized, with longer-term strategies developed to support sustainable change. Refer to Table 3 for an outline of the session content.

As enjoyment and palatability of meals is extremely important for enhancing dietary compliance, menus, recipes, and food hampers with the key components of the intervention diet were developed and disseminated to participants. Despite the complexities and detailed science underpinning the development of the diet, the ModiMedDiet intervention was designed to be easy-to-follow, sustainable, practical, and considerate of individual circumstances. Full dietary adherence was considered unrealistic.^{46,47} Instead, the primary focus was on achieving positive behaviour change and improving dietary quality through displacement of ‘discretionary items’ with healthy nutrient-dense foods. A healthy relationship with all foods was encouraged, and no foods were prohibited. By catering to individual taste preferences and the potential for appetite change, participants were more likely to be receptive to dietary advice. For example, healthy fats (e.g. olive oil and nuts), natural sugars (e.g. honey and fresh fruit), and salty foods (e.g. olives and Feta cheese), consistent with a *traditional* Mediterranean diet, were encouraged to help tackle common cravings.

Nutritional counselling component

Similar to the PREDIMED trial,³⁴ the ModiMedDiet intervention was undertaken in free-living persons; they received information, motivation, support, and empowerment from qualified dietitians with specialized nutritional counselling skills in order to modify their food habits in a real-life context, i.e. they continued to buy their own foods and cook their own meals. A study protocol manual was developed prior to trial commencement for use by the study dietitians

Table 1 ModiMedDiet food group components and example serving sizes

Food group	One serve equals	Number of servings	Other food examples
1. Wholegrains <i>Tips:</i> choose wholegrain or wholemeal varieties <i>Breakfast cereals:</i> fibre > 7.5 g per 100 g and sugar < 15 g per 100 g (< 25 g per 100 g products with dried fruit) <i>Biscuits:</i> fibre > 7.5 g per 100 g and salt < 400 mg per 100 g	1 slice bread $\frac{1}{2}$ cup cooked rice / pasta (50–60 g) $\frac{1}{2}$ cup cooked other grains (50–60 g) $\frac{1}{4}$ cup oats or muesli (30 g) $\frac{2}{3}$ cup breakfast cereal flakes (30 g) 2 Weet-Bix 3 crisp breads	5–8 per day	<i>Includes:</i> - Fibre Plus, All-bran, Weet-Bix, Porridge, Oats - Wholemeal bread or pita, wholegrain bread, sourdough bread, wholemeal English muffin, Artisan fruit toast - White flour pasta, quinoa, barley - Vita-weat biscuit, Ryvita multi-grain biscuit, Real Foods corn thins, air-popped popcorn <i>Excludes:</i> Nutri-grain, Corn-Flakes, instant noodles, white bread, raisin bread, rice crackers, rice cakes, highly processed white rice (e.g. jasmine rice, Arborio rice), gnocchi, Salada, Cruskits, water-crackers
2. Vegetables	$\frac{1}{2}$ cup cooked vegetables (75–100 g) 1 cup leafy greens (75–100 g) 1 tomato 1 small potato (75–100 g) Starchy vegetables (75–100 g)	6 per day Include tomatoes and leafy greens daily Maximum 1 potato per day	
3. Fruit <i>Tips:</i> select fresh fruit preferentially	1 medium-sized fresh fruit (150 g) 2 smaller sized fresh fruit (150 g) 1 cup tinned fruit (150 g) $1\frac{1}{2}$ tablespoons dried fruit (30 g)	3 per day Limit fruit juice to $\frac{1}{2}$ cup per day (125 mL)	
4. Dairy <i>Tips:</i> select reduced fat products and natural (or Greek) yoghurt	1 cup milk (250 mL) 200 g yoghurt 40 g hard cheese 40 g feta cheese 120 g ricotta cheese	2–3 per day	<i>Includes:</i> All dairy products. All calcium fortified soy products.
5. Nuts <i>Tips:</i> select raw and unsalted nuts – preferably walnuts and almonds	30 g nuts 30 g seeds 30 g nut spread (e.g. peanut butter) 80 g olives	1 per day	<i>Includes:</i> LSA, pumpkin seeds, sunflower seeds, etc.
6. Legumes and Pulses <i>Tips:</i> select dried or canned legumes (e.g. chickpeas, lentils, baked beans, kidney beans)	$\frac{1}{2}$ cup legumes (75–100 g) 75 g hummus 100 g tofu	3–4 per week	<i>Includes:</i> Falafel
7. Lean Red Meat <i>Tips:</i> select lean meat and remove the fat	65–100 g lean red meat $\frac{1}{2}$ cup mince 2 small chops	3–4 per week	<i>Includes:</i> lean beef, lean pork, lean ham
8. Fish <i>Tips:</i> preferentially select oily fish e.g. salmon, tuna, and sardines	100 g cooked 1 small can (95 g)	At least 2 per week Salmon 1–2 times per week (or other fish high in LC omega-3 PUFAs)	<i>Includes:</i> All fish e.g. white and oily fish. Tinned, fresh, or frozen fish. Smoked salmon, sardines, scallops, prawns, oysters, squid.
9. Poultry <i>Tips:</i> remove the skin	80–100 g poultry	2–3 per week	<i>Includes:</i> Chicken, turkey, duck
10. Eggs	1 egg (60 g)	Up to 6 eggs per week	

Continued

Table 1 Continued

Food group	One serve equals	Number of servings	Other food examples
11. Olive Oil <i>Tips:</i> select extra virgin olive oil	1TB olive oil (20 mL) 1.5TB olive oil spread/ margarine (30 g)	60 mL (3TB) per day Use daily as the main added fat	
12. Extras	1 extra portion = 120 kcal (500 kJ)	3 per week	<i>Includes:</i> Sweets, highly processed cereals, chips and savoury pastries, fried food, fast- food, fatty meat, processed meats, dairy desserts, sugary drinks, condiments
Alcohol <i>Tips:</i> select red wine preferably and only drink with meals	1 standard drink = 100 mL wine (10 g ethanol)	No more than two standard drinks per day	

(Rachelle S. Opie, intervention site # 1; Josephine Pizzinga, intervention site # 2) to ensure that a standardized dietary intervention was delivered. This protocol manual incorporated the main elements of the dietary intervention with details regarding session outline and content, delivery and dissemination of the Mod/MedDiet resource kit, nutritional counselling techniques, troubleshooting, data collection, and documentation.

With insight into the common characteristics of depression, and with awareness of the typical barriers to healthy eating faced by individuals with depressive disorders, information provision in isolation was seen as insufficient for promoting and maintaining dietary improvement. Hence, the Mod/MedDiet intervention was designed as a behaviour change intervention, with a substantial nutritional counselling component that incorporated motivational interviewing,⁴⁸ mindfulness⁴⁹ and mindful eating, and goal setting. Every effort was made to avoid a purely prescriptive approach. The dietary intervention was exclusively delivered by Accredited Practising Dietitians (Rachelle S. Opie and Josephine Pizzinga) who possessed the necessary skills, training, knowledge, educational resources, and experience to support and facilitate behaviour change. Moreover, the dietitians were able to translate the intricate science underpinning the Mod/MedDiet components into simple, clear, relevant, and individualized advice. These skills are consistent with the Role Statement for Accredited Practicing Dietitians practising in the area of Mental Health.⁵⁰

Findings from a systematic review⁵¹ provide evidence that, in addition to the beneficial role of a healthy diet, the delivery mode and interventionist (individual delivering the intervention) are likely crucial components to success of a dietary intervention. Subsequently, essential features for successful delivery of the Mod/MedDiet intervention required the dietitian to provide patient-centred care, be respectful, and instil confidence so that the patient felt empowered and capable of taking control of

their diet and other aspects of their life. Rapport building was imperative. Through developing good rapport, dietitians provided safe, comfortable, and relaxing environments⁵² conducive to the development of a trusting relationship.⁵³ The dietitian listened carefully to their patient without assumptions, bias or passing judgement and recognized the patient as the expert when it comes to information about his or her experience.⁵³ This allowed the dietitian to understand the patient's needs and concerns. Collaboration between the dietitian and patient was essential, and flexibility allowed the dietitian to tailor individual sessions to the patient's needs and interests. Dietary recommendations were considerate of individual lifestyle and social circumstances, while being culturally sensitive and amenable to potential fluctuations in mood state and motivation levels.

Incentives

To encourage dietary adherence consistent with the strategies used by the PREDIMED study,³⁴ and display the variety of foods that form the diet, participants were provided with a food hamper at the completion of the initial consult, which contained the main components of the Mod/MedDiet. A combination of fresh produce (e.g. tomatoes and lean red meat) and non-perishables (e.g. frozen berries and tinned baked beans) were included in the hamper in consideration of individuals with limited storage and cooking facilities. In an effort to support individuals with limited cooking skills/knowledge, and to assist with the incorporation of potentially unfamiliar or rarely utilized foods, recipes and meal plans using foods provided in the hamper were available to participants (Refer to Supplementary Material for details of the food hamper contents and meal plan). A second hamper was provided for attendance at the final session, to obtain final outcome measurements, and for trial completion. Moreover, it was designed to promote continued dietary adherence after intervention cessation.

Table 2 Nutrient analysis of the ModiMedDiet based on six servings of wholegrain cereals, and compared to the reference diets and the NHMRC NRVs

Nutrient	ModiMedDiet avg / day	Reference diets		NRVs National Health and Medical Research Council (NHMRC) ³⁶
		Itsipoulos et al. ³⁹	Kouris-Blazos et al. ⁴⁰	
Weight (g)	2372	—	—	—
Energy (kcal (kJ))	2596 (10 863)	2629 (11 000)	—	—
Protein (g)	113.05	—	—	RDI: 64 g/d (♂) RDI: 46 g/d (♀)
Total fat (g)	116.31	—	—	—
Saturated fat (g)	26.42	—	—	—
Polyunsaturated fat (g)	23.82	—	—	—
Monounsaturated fat (g)	58.42	—	—	—
Cholesterol (mg)	237.49	—	—	—
Carbohydrate (g)	238.13	—	—	—
Sugars (g)	106.38	—	—	—
Starch (g)	129.90	—	—	—
Water (g)	1813.43	—	—	—
Alcohol (g)	6.34	—	—	—
Dietary fibre (g)	49.88	47	—	AI: 30 g/d (♂) AI: 25 g/d (♀) RDI: 1.2 mg/d (♂) RDI: 1.1 mg/d (♀)
Thiamin (mg)	1.57	—	—	RDI: 1.3 mg/d (♂) RDI: 1.1 mg/d (♀)
Riboflavin (mg)	3.09	—	—	RDI: 16 mg/d (♂) RDI: 14 mg/d (♀)
Niacin equivalents (mg)	41.84	—	—	RDI: 45 mg/d (♂♀) AI: 5 ug/d (♂♀ 19–50) AI: 10 ug/d (♂♀ 51–70)
Vitamin C (mg)	247.52	274	—	AI: 10 mg/d (♂) AI: 7 mg/d (♀)
Vitamin D (µg)	4.38	—	—	—
Vitamin E (mg)	23.04	—	—	—
Total folate (µg)	707.41	700	—	—
Folic acid (µg)	1.98	—	—	—
Folate food (µg)	705.43	—	—	—
Folate, total DFE (µg)	708.74	—	—	RDI: 400 ug/d (♂♀)
Total vitamin A (µg) equivalents (µg)	2190.73	—	—	RDI: 900 ug/d (♂) RDI: 700 ug/d (♀)
Retinol (µg)	321.32	—	—	—
Beta carotene equivalents (µg)	11208.60	—	—	—
Sodium (mg)	1512.62	—	—	AI: 460–920 mg/d (♂♀) IUL: 2300 mg/d AI: 3800 (♂) AI: 2800 (♀)
Potassium (mg)	5724.23	—	—	RDI: 420 mg/d (♂) RDI: 320 mg/d (♀)
Magnesium (mg)	608.08	—	—	RDI: 1000 mg/d (♂♀ 19–50) RDI: 1000 mg/d (♂ 51–70) RDI: 1300 mg/d (♀ 51–70)
Calcium (mg)	1742.93	—	—	RDI: 1000 mg/d (♂♀) RDI: 8 mg/d (♂) RDI: 18 mg/d (♀ 19–50) RDI: 8 mg/d (♀ 51–70)
Phosphorus (mg)	2284.87	—	—	RDI: 14 mg/d (♂) RDI: 8 mg/d (♀)
Iron (mg)	18.00	—	—	RDI: 150 ug/d (♂♀)
Zinc (mg)	15.79	—	—	15–25% of E 20–35% of E
Iodine (µg)	159.37	—	—	—
kcal from protein	17.69% of E	12% of E	<15% of E	—
kcal from fat	39.62% of E	40% of E	40–45% of E	—
kcal from saturated fat	9.00% of E	—	10–12% of E	—
kcal from carbohydrate	36.00% of E	44% of E	38–44% of E (28% complex)	45–65% of E
kcal from sugar	16.35% of E	—	12% refined of E	—
kcal from alcohol	1.69%	4% of E from red wine	<5% of E	—
kcal from mono	20.27% of E (53.77% of total fat)	>50% of total fat	20–22% of E	—
kcal from poly	8.27% of E	—	<5% of E	—
Long Chain N3 (mg)	694.96	—	—	AI: 160 mg/d (♂) AI: 90 mg/d (♀)

Continued

Table 2 Continued

Nutrient	ModiMedDiet avg / day	Reference diets		NRVs National Health and Medical Research Council (NHMRC) ³⁶
		Itsiopoulos et al. ³⁹	Kouris-Blazos et al. ⁴⁰	
F18D2N6 (g)	19.13	—	—	AI: 13 g/d (♂) AI: 8 g/d (♀)
F18D3N3 (g)	3.74	—	—	AI: 1.3 g/d (♂) AI: 0.8 g/d (♀)
Caffeine (mg)	2.93	—	—	—
Fruits (g/day)	411	563	—	—
Vegetables (g/day)	709	691	—	—
Olive oil (ml/day)	60	75	60	—

Please note: allowing for intestinal absorption and for the nitrogenous parts of protein that cannot be completely oxidized, the average amount of energy released ranges from approximately 3.99 kcal/g (16.7 kJ/g) for carbohydrates or protein, 9.01 kcal/g (37.7 kJ/g) for fats, and 7.00 kcal/g (29.3 kJ/g) for alcohol.³⁶

Table 3 ModiMedDiet session details

Session #	Task resources	Comments
1	- Introduce the Dietitian and the intervention purpose, scope, and structure. - Assess current eating practices (complete <i>NCP – Diet History</i>). - Identify habitual behaviours, dieting and weight history, motivators, barriers, and ambivalence to change. - Assess social (e.g. family dynamics and support), environmental (e.g. food security issues, cooking facilities), physical (e.g. exercise, cooking skills, and nutrition knowledge) and medical background (e.g. mental health status, health care utilization, medication, and clinical history). - Present and discuss the <i>ModiMedDiet Food Pyramid</i>, <i>ModiMedDiet Example Serving Sizes</i>, <i>ModiMedDiet Top 10 Tips</i>. - Facilitate goal setting (complete <i>SMART Goals</i>) - Assess confidence to change (complete <i>Assessing Confidence Scale</i>). Encourage modification of goals with a confidence score of 8 or less. - Provision of food hamper (home delivered) - Display the main foods provided (<i>ModiMedDiet Food Hamper Contents</i>).	Session 1 is primarily focussed on educating the participant on the specific dietary components that comprise the ModiMedDiet. However, the dietitian must deliver this session with the knowledge that information provision in isolation does not achieve behaviour change. Dietary advice should be provided in a manner that reinforces the positive aspects of the participants' habitual eating patterns while drawing attention to areas for improvement, without being discouraging and without judgement. Make every effort to avoid overwhelming the participant. The dietitian must be highly aware of the participants' verbal and non-verbal cues when setting goals and when determining the ideal number of goals to set. Success is most likely achieved when dietary changes are compatible with, and considerate of, individual lifestyle circumstances. Where appropriate and if time permits, provide ideas and practical strategies for incorporating the foods included in the hamper.
2–6	- Review goals from previous week. - Revisit areas of particular interest or concern, or challenges faced. Facilitate discussion and problem-solving. - Facilitate goal setting (complete <i>SMART Goals</i>). - Assess whether previous goals need to be adjusted, and set new goals where appropriate. - Present and discuss any of the following topics or resource material where appropriate that reflects the participants' interests or concerns: <i>The Healthy Eating Plate</i>, <i>Convenient Meal Ideas</i>, <i>Example Meal Plan</i>, <i>Daily or Weekly Checklist</i>, <i>Shopping List</i>, <i>ModiMedDiet Recipes</i>, <i>Recipe Modification</i>, <i>Healthy Snack Ideas</i>, <i>Label Reading</i>, <i>Dining Out and Take-Away Foods</i>, <i>Mindfulness and Mindful Eating</i>, <i>Tips for Budget Buying</i>	Revisit the <i>ModiMedDiet – Top 10 Tips</i> at each session to highlight and reiterate the primary dietary objectives. Support the participant in focussing their attention to healthy, positive behaviour changes. Do not dwell on the unhealthy food choices. Instead, provide practical and individualized strategies for consuming more healthy nutrient-dense foods, which typically leads to the subsequent displacement of 'discretionary items'.
7	- Review goals from previous week. - Explore and affirm healthy behaviours achieved from this intervention. - Discuss long-term goals for achieving sustainable dietary change. - Provision of food hamper.	Reinforce the positive behaviour changes that have occurred over the intervention period to empower the individual and motivate them to continue with the desired dietary changes long term.

Assessment of diet quality

The ModiMedDiet score was specifically constructed for use in the SMILES trial by Rachelle S. Opie as a surrogate measure of diet quality, where higher scores were indicative of a better quality diet. The score allowed for measurement of adherence and compliance to the ModiMedDiet among dietary intervention participants, and was also calculated for the control group to capture any dietary changes that may have occurred as a result of being involved in the study. In brief, the ModiMedDiet score is similar to the adherence scores used in PREDIMED⁵⁴ and the Framingham Offspring Cohort⁵⁵ – a criterion-based diet score, which uses pre-defined absolute or normative goals of consumption for specific food items, independent of the individual's characteristics. Specifically, it was developed based on the recommended intakes of the 12 food group components that comprise the ModiMedDiet (Table 1), and is out of a theoretical maximum value of 120. Refer to Supplementary Material for details of the ModiMedDiet score. For reference, lowest adherence to the ModiMedDiet was defined as a score of ≤ 33.65 (Quartile 1), whereas highest adherence to the ModiMedDiet was defined as a score of ≥ 55.62 (Quartile 4).

Extras were included in the ModiMedDiet score as a dichotomous variable, as well as being reported on as a measure of diet quality independently. A lower percentage EI from *extras* (e.g. sweets, highly processed cereals, crisps, fast-food, sugary drinks) was suggestive of a healthier diet. These foods are considered unhealthy due to their high saturated fat, high salt and/or high sugar content. Refer to Table S1 for a description of foods classified as *extras*. The percentage EI from *extras* was calculated by summing the total number of kilocalories (kcal)/kilojoules (kJ) consumed as *extras* per day, and dividing this figure by the participant's average total EI. This information was obtained from FoodWorks^{7,26} based on the 7-day food diary data.

Statistical analysis

Descriptive statistics were completed first and data were checked for normality using Kolmogorov–Smirnov statistic. Continuous parametric data were expressed as mean $\pm$ SD, whereas non-parametric variables were presented as median $\pm$ IQR. Categorical variables were expressed as frequencies and percentages. The level of statistical significance was set at $P < 0.05$. All statistical analyses were performed using the SPSS program, version 20.⁵⁶ The independent samples *t*-test was used to compare the mean scores between groups of normally distributed continuous variables, whereas the Mann–Whitney U-test was employed as the non-parametric alternative.

Paired samples *t*-test was used to compare the mean scores for the groups at different time points; baseline and 3-month assessment. The Wilcoxon signed-rank test was used as the non-parametric alternative. A one-way between-groups analysis of variance was used to compare the mean scores of the dependent variable across the four groups (quartiles of ModiMedDiet score). Kruskal–Wallis test was used as the non-parametric alternative.

Results

Recruitment

Between October 2012 and July 2015, 67 individuals met all inclusion criteria and were enrolled into the SMILES trial (dietary intervention, $n = 33$; control, $n = 34$). Of these participants, four failed to submit food diaries at the baseline assessment, and a further 17 final food diaries were missing as these participants failed to complete the diary or did not attend the 3-month assessment. Hence, dietary information was available for 63 participants at baseline (dietary intervention, $n = 33$; control group, $n = 30$) and 50 participants at the final assessment (dietary intervention, $n = 26$; control group, $n = 24$).

At 3 months, the trial experienced a retention rate of 93.9% ($n = 31$) and 73.5% ($n = 25$) for the dietary intervention and control group, respectively. The 11 individuals who dropped-out of the study became uncontactable throughout the intervention period. When comparing baseline MADRS, ModiMedDiet score, anthropometric and demographic variables between 'completers' ($n = 56$) and 'drop-outs' ($n = 11$), no statistically significant differences were observed. Furthermore, participants allocated to the dietary intervention group attended significantly more sessions than those randomized to the control group (median 7 (IQR 2) versus 5 (6), respectively ($P < 0.01$)). However, session attendance did not significantly differ across quartiles of adherence to the ModiMedDiet ($P > 0.05$ NS).

Key characteristics of sample

At baseline, there were significantly more females in the diet group than in the control group ($P < 0.05$). Otherwise, groups were well matched on demographic characteristics. The mean age was 40 years (range 18–67 years), and approximately half of the individuals had a comorbid disease such as hyperthyroidism, hypertension, high cholesterol, diabetes, or arthritis.

Anthropometric measures

There were no statistically significant differences in baseline anthropometric measures for the dietary intervention and the control group ($P > 0.05$ NS). The majority (75.8%, $n = 50$) of individuals were overweight or obese. On average, females had a waist

circumference greater than 80 cm, and males had a waist circumference greater than 94 cm, which is indicative of an increased risk of chronic disease.⁵⁷ Average daily EI was 1957 kcal (8186 kJ) and 2355 kcal (9855 kJ) for females and males, respectively. Seventy-five percent of participants were considered to be 'acceptable' reporters, while 15 and 10% were under- and over-reporters, respectively.²⁷

At final assessment, no significant differences in anthropometric measures between the dietary intervention and control group were observed. Additionally, there were no significant changes over the intervention period for either group.

Diet quality

ModiMedDiet score

Despite randomized group allocation, at baseline the ModiMedDiet score was significantly lower in the dietary intervention than in the control group (36.18 versus 47.28, respectively; mean difference -11.09 , $P < 0.01$). Similarly, the percentage energy from *extras* was significantly worse (higher) in the dietary intervention than in the control group (41.73 versus 31.69, respectively ($P < 0.05$)). These differences are primarily due to lower intakes of fruit and higher intakes of *extras* in the dietary intervention group at baseline (see below).

By the end of the intervention period, the diet group tended to be more compliant with the ModiMedDiet guidelines compared to those in the control condition (ModiMedDiet score of 54.37 (SD 15.89) and 45.41 (16.03) for the diet and control group, respectively ($P > 0.05$ NS)). The effect size for this difference was a Cohen's d of 0.56. More specifically, individuals in the diet group had improved their ModiMedDiet score by an average of 19.88 points (mean 88.0% score increase), and the percentage of energy from *extras* had declined by a mean of 17.47 points. Conversely, on average, the ModiMedDiet score and the percentage of energy from *extras* had not improved in the control group – a mean change in ModiMedDiet score of -1.87 points, and a mean increase in *extras* of 1.26 points.

ModiMedDiet food group components

At baseline, there were no significant differences in intake between the dietary intervention and control group for the ModiMedDiet food group components, with the exception of fruit and *extras* (Table 4). By final assessment, the diet group were consuming significantly more olive oil ($P < 0.001$) than the social support control group, and there was a trend for a higher intake of vegetables and whole grain cereals (Table 4).

At intervention cessation, the diet group were shown to have significant improvements in the consumption

of the following food groups from baseline; whole-grain cereals (mean increase 1.21 (SD 1.77) servings/day, $P < 0.01$), fruit (0.46 (0.71) servings/day, $P < 0.01$), dairy (0.52 (0.72) servings/day, $P < 0.001$), olive oil (0.42 (0.49) servings/day, $P < 0.001$), pulses (1.40 (2.39) servings/week, $P < 0.01$), and fish (1.12 (2.65) servings/week, $P < 0.05$) (Table 4). Moreover, the magnitude of dietary changes observed in this trial for whole grain cereals, vegetables, dairy, pulses, and fish was greater than those seen in the seminal PREDIMED study.⁵⁸ Hence, implementation of the ModiMedDiet is considered to be of high acceptability and feasibility in a non-Mediterranean population. With respect to the consumption of unhealthy food items, intake of *extras* substantially declined in the diet group (a mean decrease of 21.76 serves per week ($P < 0.001$)). Conversely, there were no significant improvements from baseline observed in the social support control group for all 12 ModiMedDiet food group components (Table 4).

Nutrient intake

By final assessment, the dietary intake of protein (as a % of EI), polyunsaturated fat (as a % of EI), dietary fibre, vitamin E, long chain (LC) omega-3 PUFAs, and α -linolenic acid had all significantly increased among the diet group (Table 5). This is consistent with improvements in intake of food groups associated with the ModiMedDiet (as noted above). Additionally, carbohydrate intake had significantly decreased by a mean value of 29.6 grams, and as a percentage of total energy (from 44.1 to 40.7%). Conversely, diet quality as measured by nutrient density deteriorated in the control group. Specifically, the intake of energy, thiamin, vitamin E, β carotene equivalents, potassium, iron, and α -linolenic acid all statistically significantly declined from baseline to final assessment in the control condition (Table 5).

Discussion

This paper describes in detail the ModiMedDiet intervention, which is, to our knowledge, the first prescriptive, individualized dietary intervention specifically designed for adults with major depression. The information provided in this manuscript provides a toolkit for dietitians, and is designed to aid dietitians and researchers in the delivery of the ModiMedDiet to individuals with depression, in order to achieve improvements in diet quality and depressive symptoms as observed in our trial.²¹

Specifically, the results of this study demonstrate that with nutrition counselling support from a qualified dietitian, improvements in dietary quality can be achieved in a population with MDD. The diet group were observed to be more compliant with the ModiMedDiet guidelines, which were designed

Table 4 ModiMedDiet food group components at baseline and 3-month assessment

ModiMedDiet food group component	Time point	Diet group mean (SD) <i>n</i> = 33 baseline <i>n</i> = 26 final		Control group mean (SD) <i>n</i> = 30 baseline <i>n</i> = 24 final		<i>P</i> ^b (<i>t</i> , <i>df</i>)	<i>P</i> ^c (<i>t</i> , <i>df</i>)	
							Diet	Control
<i>Serves/Day</i>								
Wholegrain cereals	<i>Baseline</i>	2.08 (1.51)	Mean Change 1.21 (1.77)	2.74 (1.61)	Mean Change −0.42 (1.34)	0.098 (−1.68, 61)	0.002 (3.47, 25)	0.138 (−1.54, 23)
	<i>Final</i>	3.24 (1.64)		2.37 (1.81)		0.081 (1.79, 48)		
Vegetables	<i>Baseline</i>	1.73 (0.89)	0.53 (1.25)	1.91 (1.28)	−0.14 (1.16)	0.495 (−0.69, 61)	0.058	0.577
	<i>Final</i> ^a	2.05 (1.63)		1.53 (1.08)		0.062		
Fruit	<i>Baseline</i> ^a	0.60 (0.75)	0.46 (0.71)	1.00 (1.51)	−0.05 (0.80)	0.035	0.004	0.989
	<i>Final</i>	1.10 (0.80)		1.26 (0.88)		0.489 (−0.70, 48)		
	<i>Baseline</i>	1.19 (0.84)		0.52 (0.72)	1.57 (0.95)	−0.15 (0.82)		
Nuts	<i>Final</i>	1.59 (0.74)		1.36 (0.79)		0.309 (1.03, 48)		
	<i>Baseline</i> ^a	0.00 (0.60)	0.13 (0.56)	0.17 (0.63)	0.13 (0.83)	0.187	0.256	0.673
	<i>Final</i> ^a	0.44 (0.65)		0.16 (0.71)		0.440		
Olive Oil	<i>Baseline</i> ^a	0.00 (0.20)	0.42 (0.49)	0.00 (0.25)	−0.10 (0.24)	0.750	0.000	0.037
	<i>Final</i> ^a	0.25 (0.76)		0.00 (0.08)		0.000		
	<i>Baseline</i> ^a	0.00 (0.60)		1.40 (2.39)	0.12 (4.55)	−0.70 (3.57)		
Pulses	<i>Final</i> ^a	0.18 (2.97)		0.00 (2.33)		0.198		
	<i>Baseline</i> ^a	3.06 (3.77)	−0.26 (3.06)	3.21 (4.61)	−0.38 (3.33)	0.572	0.681	0.693
Red Meat	<i>Final</i>	2.82 (1.77)		2.50 (2.35)		0.587 (0.55, 48)		
	<i>Baseline</i> ^a	0.44 (1.80)	1.12 (2.65)	1.38 (2.50)	0.23 (2.16)	0.121	0.042	0.852
	<i>Final</i> ^a	1.00 (3.04)		1.10 (2.95)		0.896		
Chicken	<i>Baseline</i> ^a	3.00 (5.23)	−0.14 (4.26)	1.94 (2.99)	0.01 (3.34)	0.104	0.757	0.853
	<i>Final</i> ^a	2.38 (3.08)		2.00 (2.63)		0.565		
	<i>Baseline</i> ^a	1.00 (2.00)	0.40 (1.83)	1.88 (3.13)	0.12 (2.19)	0.077	0.297	0.722
Eggs	<i>Final</i> ^a	2.00 (3.00)		1.08 (4.00)		0.984		
	<i>Baseline</i> ^a	49.62 (38.64)	−21.76 (16.01)	39.44 (33.85)	−1.36 (16.63)	0.052	0.000	0.786
	<i>Final</i> ^a	20.45 (34.66)		35.66 (36.03)		0.361		

Bold values indicate a statistically significant finding ($p < 0.05$).

^aNon-parametric data presented as median (IQR).

^b*P*-values (*t*, *df*) for independent samples *t*-test (or Mann–Whitney U-test for non-parametric data).

^c*P*-values (*t*, *df*) for paired samples *t*-test (or Wilcoxon signed-rank test for non-parametric data).

Table 5 Macro- and micro-nutrient intake at baseline and final assessment

Nutrient	Time point	Diet group (n = 33 baseline; n = 26 final) Mean (SD)	Control group (n = 29 baseline; n = 24 final)	<i>P</i> ^b (<i>t</i> , <i>df</i>)	<i>P</i> ^c (<i>t</i> , <i>df</i>)	
					Diet	Control
Average Energy (kcal)	Baseline	2061 (632)	2057 (438)	0.980 (0.03, 60)	0.355 (−0.94, 25)	0.023 (−2.45, 22)
	Final	1971 (491)	1878 (423)	0.478 (0.72, 48)		
Protein (% of E)	Baseline	17.46 (3.05)	17.70 (2.96)	0.754 (−0.32, 60)	0.009 (2.83, 25)	0.683 (−0.41, 22)
	Final	18.15 (3.09)	16.98 (3.18)	0.193 (1.32, 48)		
Total fat (% of E)	Baseline	35.27 (5.64)	34.44 (5.43)	0.558 (0.59, 60)	0.212 (1.28, 25)	0.400 (0.86, 22)
	Final	37.20 (6.34)	34.98 (7.01)	0.245 (1.18, 48)		
Sat Fat (% of E)	Baseline	14.47 (3.54)	13.53 (2.88)	0.258 (1.14, 60)	0.500 (−0.68, 25)	0.201 (1.32, 22)
	Final	14.21 (3.58)	14.40 (3.73)	0.858 (−0.18, 48)		
Poly fat (% of E)	Baseline	5.01 (1.68)	5.46 (1.73)	0.304 (−1.04, 60)	0.017	0.951
	Final ^a	5.72 (3.04)	5.29 (1.15)	0.290		
Mono fat (% of E)	Baseline	13.14 (2.56)	12.64 (2.58)	0.445 (0.77, 60)	0.091	0.858
	Final ^a	14.15 (4.35)	11.68 (2.73)	0.015		
Mono fat (% of Fat)	Baseline	37.33 (4.52)	36.73 (4.44)	0.600 (0.53, 60)	0.234 (1.22, 25)	0.256 (−1.17, 22)
	Final	38.40 (5.26)	34.80 (4.60)	0.013 (2.57, 48)		
Carbohydrate (% of E)	Baseline	44.10 (6.42)	44.63 (6.62)	0.749 (−0.32, 60)	0.003 (−3.34, 25)	0.611 (−0.52, 22)
	Final	40.70 (6.04)	45.10 (8.01)	0.033 (−2.20, 48)		
Sugars (% of E)	Baseline	20.02 (5.27)	20.57 (5.59)	0.688 (−0.40, 60)	0.159 (−1.45, 25)	0.703 (0.39, 22)
	Final	17.96 (5.47)	21.82 (6.77)	0.031 (−2.23, 48)		
Dietary fibre (g)	Baseline	21.33 (6.27)	24.91 (9.02)	0.072 (−1.83, 60)	0.025	0.074
	Final ^a	24.39 (9.19)	19.98 (9.58)	0.236		
Thiamin (mg)	Baseline ^a	1.57 (1.02)	1.71 (1.01)	0.516	0.780	0.028
	Final ^a	1.60 (0.91)	1.36 (1.07)	0.472		
Niacin equiv. (mg)	Baseline ^a	45.63 (15.11)	45.75 (17.78)	0.657	0.790	0.059
	Final ^a	42.94 (16.15)	38.37 (17.52)	0.372		
Vitamin E (mg)	Baseline	7.80 (2.86)	9.70 (4.00)	0.034 (−2.17, 60)	0.026 (2.36, 25)	0.006 (−3.06, 22)
	Final	10.25 (5.07)	7.58 (3.07)	0.031 (2.23, 48)		
Vitamin A equiv. (µg)	Baseline ^a	655.13 (552.65)	606.35 (468.39)	0.816	0.137	0.024
	Final	882.10 (476.73)	613.62 (205.28)	0.013 (2.62, 35)		
B carotene equiv. (µg)	Baseline ^a	2079.47 (2928.79)	2277.76 (2486.46)	0.893	0.058	0.036
	Final ^a	2816.92 (3592.27)	1783.73 (1648.62)	0.021		
Potassium (mg)	Baseline ^a	2768.52 (1112.38)	2880.05 (856.89)	0.494	0.144	0.042
	Final ^a	2781.36 (1077.97)	2616.96 (1357.54)	0.382		
Iron (mg)	Baseline ^a	10.92 (5.31)	12.48 (5.01)	0.320	0.949	0.008
	Final ^a	11.42 (4.86)	10.84 (4.57)	0.567		
LC n-3 (mg)	Baseline ^a	107.47 (155.75)	215.23 (334.95)	0.043	0.015	0.855
	Final ^a	151.54 (315.91)	181.78 (297.08)	0.712		
α-Linolenic acid (g)	Baseline ^a	1.07 (0.70)	1.17 (0.56)	0.182	0.018	0.048
	Final ^a	1.27 (1.60)	1.02 (0.43)	0.014		

Bold values indicate a statistically significant finding ($p < 0.05$).

^aNon-parametric data presented as median (IQR).

^b*P*-values (*t*, *df*) for independent samples *t*-test (or Mann–Whitney U-test for non-parametric data) to compare difference between Mod/MedDiet intervention and control group.

^c*P*-values (*t*, *df*) for paired samples *t*-test (or Wilcoxon signed-rank test for non-parametric data) to compare the groups at different time points (baseline and 3-month assessment).

specifically for the SMILES trial and based on existing dietary guidelines and recommendations,⁴⁴ as well as scientific evidence from the emerging field of nutritional psychiatric epidemiology.

Although individuals with depressive symptoms are perceived to face many barriers to healthy eating, the magnitude of dietary changes, observed among diet group participants for mean intake of whole grain cereals, vegetables, dairy, pulses, and fish, was greater than that seen in the seminal PREDIMED study.⁵⁸ Such differences are potentially explained by the fact that participants enrolled in the SMILES trial had poor quality diets at baseline,²⁵ which were generalizable to the Australian population,⁵⁹ whereas the participants recruited for PREDIMED were already following a 'reasonably good' Mediterranean-style food pattern.⁵⁸ Nonetheless, our findings demonstrate that the ModiMedDiet intervention, can support individuals with MDD to overcome the many barriers to healthy eating to achieve dietary improvement. Feedback from participants regarding how they felt after following the diet were overwhelmingly positive and indicative of the high acceptability of the dietary intervention; 'I feel great and I feel confident I can continue to make dietary choices that make me feel good physically AND emotionally'; 'really indescribably good! On some intricate, internal level my body felt much better. I feel like my brain functions with more clarity. I have more energy'.

A number of intervention components are likely to have supported enhanced dietary compliance, such as the free provision of a food hamper, the individualized approach, goal-setting, and the nutritional counselling (e.g. motivational interviewing and mindfulness). Dietary adherence is also likely to have been enhanced by the dissemination of shopping lists, meal plans, and recipes.⁵⁸ Unfortunately, this study was unable to provide evidence to indicate that these beneficial changes will be sustained beyond the 3-month intervention period when participants no longer receive advice or support from a dietitian. However, when considering that participants were provided with practical strategies for purchasing, preparing and cooking these highly palatable, enjoyable, and satisfying foods⁵⁸ in real-life conditions, it is highly plausible that these dietary changes will be maintained. Sustainability of a Mediterranean-type diet has been demonstrated in the PREDIMED longitudinal study, but has yet to be shown in a non-European cohort. This RCT has successfully demonstrated that a Mediterranean-style diet can be effectively implemented in an Australian population.

The dietary information utilized in this analysis was obtained from self-reported intake of foods and beverages using 7-day food diaries, commonly referred to as the 'gold standard' measure.⁶⁰

Moreover, the data were examined by an RA or dietitian for accuracy of reporting, and under- and over-reporting²⁷ were calculated to provide an indicator of the likely validity of the food diary information. The food diaries included detailed information about food choices not available from a food frequency questionnaire. However, the subjective nature of self-reported dietary intake assessment methods presents numerous challenges to obtaining accurate dietary intake and nutritional status. This limitation can be overcome by the use of dietary biomarkers, which are able to objectively assess dietary consumption (or exposure) without the potential bias of self-reported dietary intake errors.⁶¹ Carotenoids, as biomarkers of fruit and vegetable intakes, will be presented in future manuscripts.

Future recommendations

A key feature of this programme involved the intervention delivery by Accredited Practising Dietitians. To promote fidelity, quality, and future scalability within the context of existing health services, funding mechanisms that allow qualified dietitians to be subsidized for this activity with clinically depressed populations would be advantageous. For example, in Australia, to allow for better access to dietitians, consideration should be made to include dietitians as part of the GP Mental Health Treatment Plan.⁶² Creating a Medicare item number for dietitians on the mental health treatment plan would limit the out-of-pocket costs incurred, and allow for Medicare rebates of up to ten individual and ten-group sessions per calendar year to patients with an assessed mental disorder.

Acknowledging that dietetic services to individuals with mental health issues are currently limited, other more accessible mediums for delivering the ModiMedDiet should be considered. This is especially pertinent when considering that individuals are unlikely to achieve lifestyle and behaviour changes without support from health professionals. Hence, a hybrid approach, which uses a combination of digital platforms (e.g. interactive-web based programs, e-health initiatives, and social media) delivered by qualified dietitians, may be the best option. This approach can reach large population numbers at relatively low cost, enhance patient accessibility, and reduce patient and clinician burden. The digital technology platform also addresses multiple key functions of effective health behaviour change models, e.g. education provision, health coaching,⁶³ monitoring, and social support.⁶⁴

Conclusion

The SMILES RCT is the first study of its kind, which employed a qualified dietitian to develop and deliver an individually tailored ModiMedDiet to individuals

with major depression in order to improve diet quality. An unhealthy diet is one of the leading risk factors for the non-communicable diseases commonly comorbid with depression,⁶⁵ whereas this nutrition counselling intervention was successful at improving the dietary habits of diet group participants. Thus, implementation and execution of a dietary counselling intervention, such as the ModiMedDiet, should be considered as a potentially important therapeutic approach for the treatment of depression,²¹ which is also likely to yield a range of health benefits.

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Ethics approval None.

Supplementary material

The underlying research materials for this article (Supplemental data) can be accessed at 10.1080/1028415X.2017.1312841.

ORCID

Felice N. Jacka  <http://orcid.org/0000-0002-9825-0328>

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